Kaduna.lk

A Unified Smart Platform for Roadside Assistance and Automotive Services in Sri Lanka

R26-SE-026 | Project Proposal Defense

Team Members



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The Problem

- **Sri Lanka has NO dedicated digital platform for roadside assistance**
- Drivers rely on manual phone calls → average wait: 45–90 minutes in urban areas
- No mechanism to assess traffic severity or congestion, or prioritize dispatch
- Insurance claims vulnerable to fraud — 2D photos lack depth verification
- Reactive vehicle maintenance → unexpected breakdowns, accidents, higher costs

A Unified Smart Platform for Roadside Assistance and Automotive Services in Sri Lanka



Users, Stakeholders & Impact

- **Key Stakeholders:**
  - Stranded Drivers — daily commuters, commercial operators, elderly/women drivers
  - Roadside Assistance Providers — tow trucks, mechanics, fuel delivery
  - Insurance Companies — 14 licensed providers in Sri Lanka
  - Traffic Authorities — congestion monitoring and infrastructure planning
  - Vehicle Service Centers — 181+ registered repair centres

- **Impact on Society:**
  - Prolonged wait times → driver stress, safety risks, economic losses
  - Cascading traffic congestion → fuel wastage, emissions, productivity loss
  - Mismatched providers → wasted trips (25–35% mismatch rate currently)
  - Insurance fraud → financial losses for companies and policyholders

Existing Solutions & Their Limitations

- **Current State in Sri Lanka:**
  - Manual phone-based coordination — no digital dispatch system exists
  - Insurance apps — simple 2D photo upload, vulnerable to claim recycling
  - Reactive maintenance — repairs only after failure, no prediction

- **International Platforms:**
  - Honk / Urgently (USA) — deterministic nearest-match dispatch
    - Limitation: Assume service type is known → high mismatch when diagnosis is uncertain
    - No integration with vehicle telemetry or traffic impact data

- **No system globally combines: dispatch optimization + traffic intelligence + predictive maintenance + privacy compliance**

Proposed Solution: Kaduna.lk Platform

A unified smart platform with 4 integrated research components:

1. Dispatch Optimization & Smart Recovery — Janukshan Sivakumar (IT22635266)
2. Geo-Intelligence & Traffic Impact Analysis — Mohamed Asath (IT22633422)
3. Predictive Maintenance for Service Management — Samudith Tharindaka (IT22639776)
4. Intelligent 3D Accident Claim System— Dilnuk De Silva (IT22001252)

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Component: Dispatch Optimization & Smart Recovery

Contribution

Core decision making engine — determines what service an incident actually needs and selects the optimal service provider/ assistance minimizing mismatch rates and traffic congestion.

Novelty

First framework to treat roadside dispatch with **service type uncertainty**, considering external traffic and cost for dispatch optimization and adaptive learning.

Dispatch Optimization & Smart Recovery

Problem

Drivers rely on Manual phone calls for roadside dispatch and find difficult to connect to the right service provider for their need and may end up in wrong service provider, which costs time and energy.

Research Gap

- No digital dispatch system for roadside assistance in Sri Lanka.
- No Dispatch Optimization if the Issue is uncertain/unknown.
- No vehicle telemetry in dispatch decisions.
- No Externality-Aware Dispatch for Roadside Assistance.
- No Adaptive Learning from Resolution Outcomes

Objectives

- Multi-source Diagnostic Triage
- Expected-Cost Minimization algorithm selecting providers under uncertainty
- Adaptive Bayesian Learning updating diagnostic priors from outcomes
- Simulation Evaluation with expected outcomes.

Dispatch Optimization & Smart Recovery

Uncertainty-Aware Dispatch Optimization (UADO) framework

- Stage 1 — Diagnostic Triage System
- Stage 2 — Expected-Cost Minimization (ECM)
- Stage 3 — Adaptive Bayesian Learning

Latest Technologies Used to Develop the Proposed Solution

- **Stochastic Optimization** ECM Algorithm evaluates providers across all possible service outcomes weighted by probability
- **Bayesian Inference**: Continuously updates diagnostic probabilities after each incident using Bayes' theorem.
- **Decision Tree Triage**: A structured 7-question diagnostic flow, combined with real-time OBD-II telemetry, produces probability distributions over 9 service categories.
- **Multi-Source Data Fusion**: Merges user-reported symptoms, OBD-II fault codes, sensor data, traffic impact scores, and provider trust scores into a unified dispatch decision.
- **Discrete Event Simulation** (SimPy)

Dispatch Optimization & Smart Recovery

Validation & Evaluation

Discrete event simulation: 1,000 synthetic incidents across Colombo over 14 days, 50 service providers. Compared against 3 baselines: Greedy-Nearest, Type-Matched Nearest, Deterministic Multi-Criteria.

Metrics: Mismatch Rate, Diagnostic Accuracy, Mean resolution Time

Data Availability, Security & Ethics

Data: Datasets available for structured fault records with symptoms and Dataset for engine fault classification.

Security: PDPA-compliant design with data minimization, explicit consent for OBD-II access, anonymized incident storage, JWT authentication, TLS 1.3 encryption.

Ethics: Synthetic data only during research phase. No human subjects involved. Privacy-by-design principles applied throughout.

Dispatch Optimization & Smart Recovery

System Overview

Internal Architecture: 4 components working together:

1. Incident Manager - orchestrates full lifecycle
2. Diagnostic Triage Engine - processes questionnaire + OBD data
3. Dispatch Optimizer - runs ECM algorithm
4. Bayesian Learning Engine - updates diagnostic priors

Dependencies & Integration

- From Predictive Maintenance: OBD-II data (battery voltage, fault codes, vehicle health)
- From Geo-Intelligence: Traffic impact scores (1-10 scale)



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Component: Geo-Intelligence & Traffic Impact Analysis

Contribution: Spatial awareness layer — provides traffic intelligence to dispatch and congestion visualization to traffic authorities

Novelty: First incident impact scoring model calibrated for Sri Lankan road conditions, validated against SUMO microscopic traffic simulation

Geo-Intelligence & Traffic Impact Analysis

- **Research Gap & Novelty**
  - No traffic impact scoring model exists for Sri Lankan roads
  - 73% of roads are 2-lane (high vulnerability) | 78% residential (limited alternatives)
  - Dispatch systems lack traffic intelligence — treat all incidents equally
  - No spatial hotspot analysis of vehicle incidents in Sri Lanka
  - Google Maps shows current traffic — cannot predict future congestion from a new incident
- **Novel Contributions:**
  - 5-factor Impact Scoring Model (1–10): CLF, TVF, TF, LF, ISF
  - SUMO-validated scoring (target: Pearson r > 0.75)
  - Impact-weighted hotspot analysis (frequency × severity)
  - Congestion prediction using LWR shockwave theory
- **Objectives: Scoring model → SUMO validation → Hotspot analysis → Dashboard → Priority dispatch (≥20% VHL reduction)**

Geo-Intelligence and Traffic Impact Analysis— Methodology

- **6-Phase Methodology:**
- Phase 1: Data Collection — Colombo road network via OSMnx (42,989 nodes, 96,881 edges)
- Phase 2: Impact Scoring — Score = $(w_1 \times CLF + w_2 \times TVF + w_3 \times TF + w_4 \times LF + w_5 \times ISF) \times 10$
  - Priority: CRITICAL (8–10), HIGH (5–7.9), MEDIUM (3–4.9), LOW (1–2.9)
- Phase 3: SUMO Simulation — 90 scenarios (5 road types × 2 lanes × 3 volumes × 3 durations)
- Phase 4: Weight Optimization — scipy SLSQP + ML comparison (RF, GB)
- Phase 5: Hotspot Analysis — KDE density surface + DBSCAN clustering
  - Composite risk = Incident Count × Average Impact Score
- Phase 6: Dashboard — Next.js + Leaflet.js with heatmap + what-if simulator
- **Validation Targets:**
  - Pearson r > 0.75 | MAE < 1.5 | Queue error < 20% | Silhouette > 0.3 | Load < 3s

Geo-Intelligence and Traffic Impact Analysis — Implementation

- **3-Layer Architecture:**
  - Data Layer: Road network DB (OSM), traffic volume estimates, incident data
  - Analysis Engine: Impact Scoring + Congestion Prediction + Hotspot Analysis + What-If Simulator
  - Output Layer: Priority Signal API (JSON) + Dashboard + Reports
- **Integration:**
  - → Dispatch (Janukshan): Provides impact scores + priority levels via REST API
  - ← Dispatch: Receives dispatch timestamps + ETAs for dashboard updates
  - Stateless scoring function → horizontally scalable behind load balancer
- Tech Stack: Python 3.12+, OSMnx, SUMO, SciPy, scikit-learn, Next.js 16, Leaflet.js, Tailwind CSS

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Component Predictive Vehicle Maintenance

Contribution

An AI-driven system that predicts vehicle component failures before they happen and automatically connects users to an integrated marketplace for timely parts replacement and professional maintenance services.

Novelty

A smart system that uses engine data, driving style, and history to predict vehicle breakdowns and suggest the right parts and services in the intergrated marketplace

Predictive Vehicle Maintenance

Problem

Current vehicle maintenance systems often ignore past service records, do not connect failure predictions to parts or garage bookings, and are not designed for older vehicles or local service conditions.

Research Gap

- **Data Omission**: Ignore past service records
- **Disconnected Workflow:** Predictions not linked to parts or garages
- **Local Mismatch:** Designed for new cars, not older vehicles or local mechanics

Objective

- Integrate data: OBD-II + driver behavior + service history
- Predict health & RUL of vehicle components
- Optimize maintenance scheduling to reduce costs
- Connect predictions to marketplace for parts and service booking

Latest Technologies Used to Develop the Proposed Solution

- **Machine Learning:** Random Forest and Gradient Boosting models using Scikit-learn in Python for Remaining Useful Life (RUL) prediction.
- **Optimization:** PuLP for selecting the best repair option based on cost, distance, and availability.
- **Simulation:** Simulation of Urban Mobility for testing the system under 50 simulated driving scenarios.

Predictive Vehicle Maintenance

Validation & Evaluation

- Accuracy: Verified AI predictions with MAE, RMSE, and R^2 using 5-fold cross-validation.
- Efficiency: Measured cost and downtime reduction versus traditional schedules.
- Reliability: Stress-tested closed-loop responses in 50 behavior-driven simulations.

Data Availability, Security & Ethics

Data:

- Open source: online resources like zenodo and kaggle
- SUMO simulation : 50 Scenarios (5 Driver Profiles × 5 Traffic Densities × 2 Network Types)
- Manual collection for service records

Security: PDPA-compliant design with data minimization, explicit consent for OBD-II access, JWT authentication, TLS 1.3 encryption.

Ethics: Synthetic data only during research phase. No human subjects involved. Privacy-by-design principles applied throughout.

Predictive Vehicle Maintenance

System Architecture



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Component Intelligent 3D Accident Claim System

Contribution: A mobile-first 3D accident scene reconstruction system using guided photo capture, low-light enhancement, and volumetric fraud detection to replace unreliable 2D photo submissions.

Intelligent 3D Accident Claim System

Problem

- Claim recycling - old damage photos submitted as new claims, no depth verification.
- Nighttime accidents - dark photos unusable, forcing slow manual inspections.
- No depth perception - 2D photos cannot capture true damage depth or severity.

Research Gap

- Damage assessment only, no 3D reconstruction.
- 3D reconstruction, no low-light or fraud validation.
- Fraud detection, no visual or spatial verification.
- No existing study combines all three in a single pipeline.
- No Sri Lankan insurance system addresses these gaps together.

Novel Contributions

- 3D + 360° scene reconstruction from mobile photos
- Low-light enhancement
- Temporal metadata anchoring
- Guided photo capture, all in one system.

Intelligent 3D Accident Claim System

Research Objectives

- Guided Capture — minimum 15 photos, with 70% overlap, using sensors-locked.
- Low-Light Enhancement — Zero-DCE restores nighttime photos on-device.
- 3D Reconstruction — textured mesh via OpenMVG and OpenMVS.
- Fraud Validation — GPS, timestamp, and depth cross-check.

Intelligent 3D Accident Claim System

- 5-Stage Methodology
  - Stage 1: Guided Capture: 15 photos, 70% overlap, gyroscope-locked shutter.
  - Stage 2: Low-Light Enhancement: Zero-DCE on-device, under 80KB model.
  - Stage 3: 3D Reconstruction: OpenMVG → OpenMVS → Poisson mesh → textured OBJ, under 6 min.
  - Stage 4: Fraud Validation: GPS + timestamp cross-check, depth matched against impact physics.
  - Stage 5: Role-Scoped Delivery: Owner → status only. Insurer → full mesh. Garage → damage report.

Validation Targets

- 30 vehicles — 10 minor, 10 moderate, 10 structural.
- MAE target < 10mm vs licensed inspector ground truth.
- BRISQUE score improvement ≥ 15 points for low-light enhancement.
- Surface coverage target ≥ 85% for nighttime captures.

Intelligent 3D Accident Claim System

Architecture

- 3D Reconstruction - SfM pipeline runs asynchronously on a cloud server.
- Job Queue - app receives job ID instantly, push notification when the model is ready.
- Encrypted Storage - AES-256 for all 3D models and claim data at rest.
- Secure Transmission - TLS 1.3 for all data in transit.
- Gallery uploads disabled - only in-app camera captures accepted.

Integration:

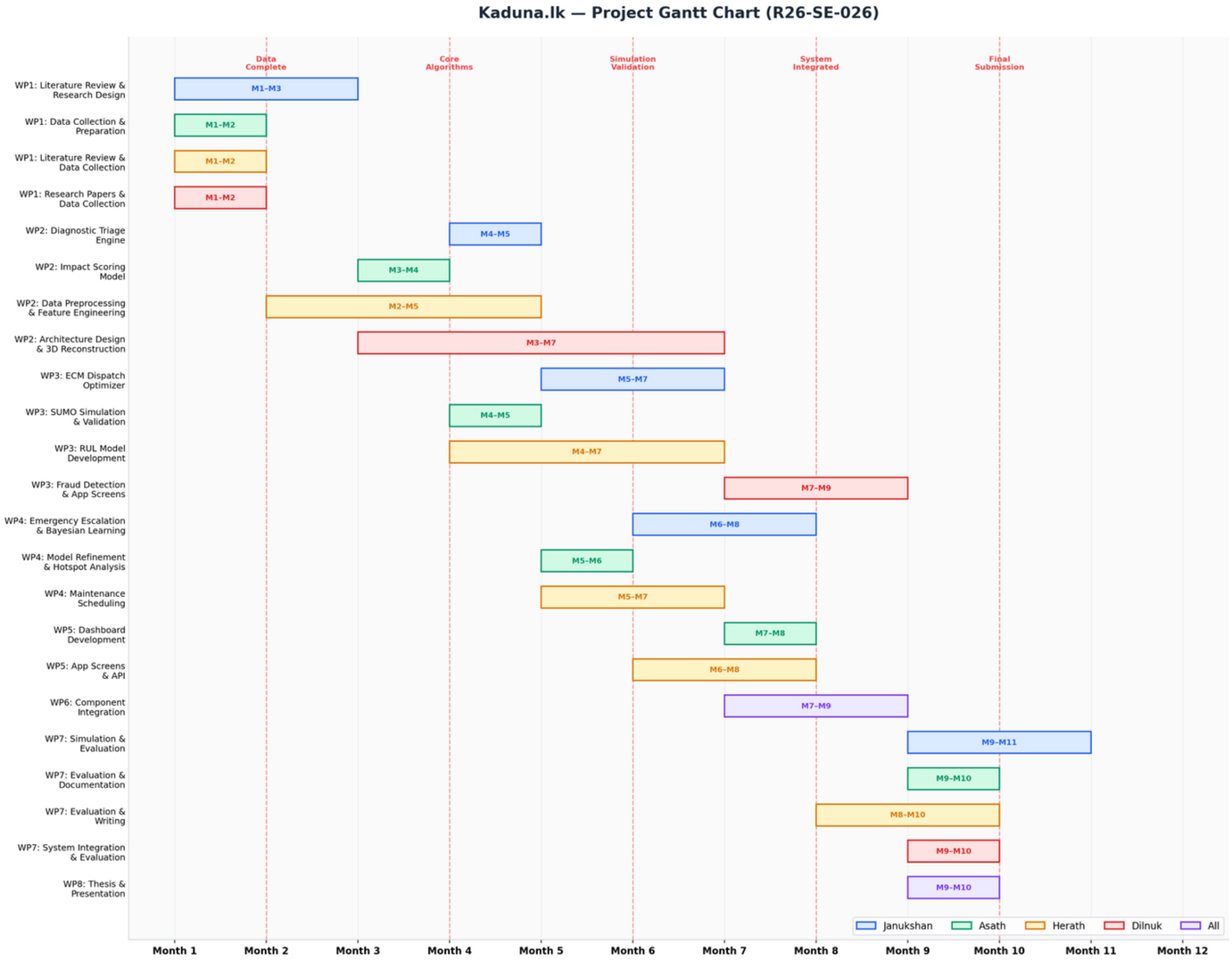
- Mobile app captures photos → uploads to reconstruction service → 3D model stored → masked output delivered per role.

Tech Stack : React Native · OpenMVG · OpenMVS · Zero-DCE · Azure Service Bus · Azure Blob Storage · JWT + AES-256
Python · Node.js · Three.js

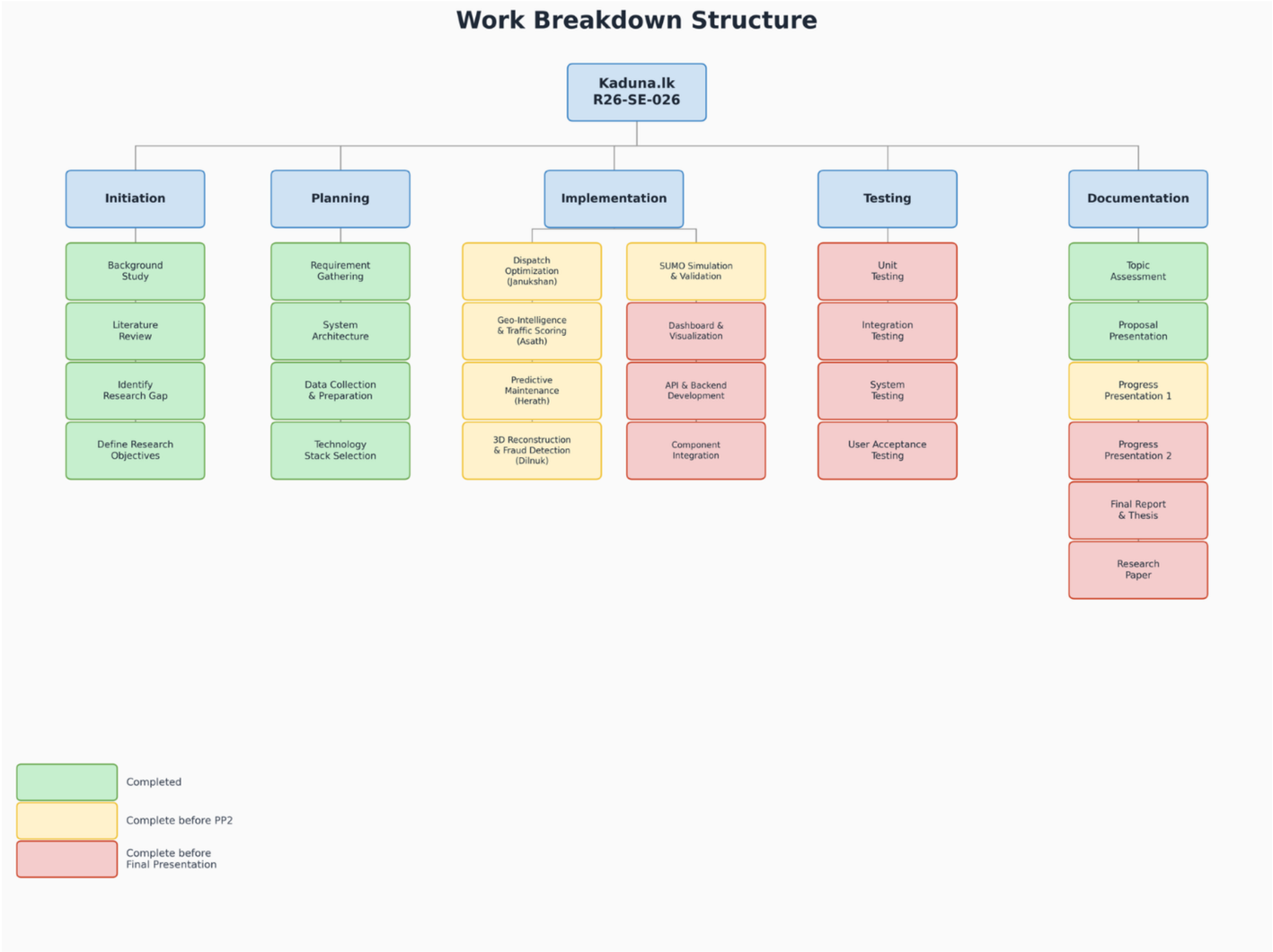
Commercialization Plan

- **Target Market:**
  - Private vehicle owners, Fleet operators, insurance companies, repair centres, Traffic authorities
- **Value Proposition:**
  - First unified smart platform for roadside assistance in Sri Lanka — intelligent dispatch, traffic intelligence, predictive maintenance, 3D fraud prevention, PDPA compliance
- **Budget:**
  - Software Costs and Marketing - LKR 80,000–120,000
- **Revenue Model:**
  - Freemium for individual drivers, B2B subscription for fleet/insurance, Government licensing, Marketplace commission, Advertising
- **Competitive Advantage:**
  - First ever in Sri Lankan market
  - Validated through simulation & Integrated platform approach
- **Intellectual Property Rights**:
  - Proprietary algorithms eligible for protection - Academic publications planned
  - Software Copyrights and Patent
  - Trademark for Branding

Project Timeline



Work Breakdown Structure



Thank You

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Questions & Discussion